

Accelerating Design Exploration and Optimization with Surrogate Physics Models

AN END SEMESTER PROJECT REPORT
SUBMITTED IN PARTIAL FULFILMENT OF THE
REQUIREMENTS FOR THE DEGREE OF
Bachelor of Technology in Engineering and Computational Mechanics

by

Shiv Ratn (2022AM11816)
Rishabh Jain (2022AM11793)

Under the guidance of

Prof. Rajdip Nayek
Prof. Sanjeev Sanghi (Co-supervisor)



Applied Mechanics
Indian Institute of Technology, Delhi
Hauz Khas
New Delhi– 110 016

November 2025

Acknowledgements

We wish to express our sincere gratitude to our project supervisors, Prof. Rajdip Nayek and Prof. Sanjeev Sanghi, for their guidance, insightful feedback, and unwavering support throughout this B.Tech. Project. Their expertise was invaluable in shaping the direction of this research.

Abstract

This report develops a surrogate-assisted optimisation framework for the inverse design of a gyroid heat exchanger. The goal is to maximise surface area subject to strict limits on mass, pressure drop, and mean flow velocity. The framework is built around two coupled Gaussian Process Regression (GPR) models that separate geometry-related outputs from flow predictions, allowing the surrogate to provide both accurate estimates and useful uncertainty information.

Training data was generated through a hybrid strategy. An initial 75-point Latin Hypercube Sampling (LHS) sweep defined the broad design space, and an Active Learning loop then added new samples in regions close to the 125 g mass boundary or where the surrogate uncertainty was high. This kept the dataset compact while improving model reliability in the most relevant regions.

The optimisation process used the surrogate in two stages. A global Bayesian search identified the promising part of the design space, followed by a trust-region refinement that added 50 local LHS samples and retrained the surrogate before running a final optimisation step. The method produced a design that maximised surface area while meeting the mass constraint exactly, reducing what would ordinarily require weeks of simulation time to an effort manageable within hours.

Contents

Acknowledgements	i
Abstract	ii
Notation and Abbreviations	vii
1 Introduction	1
1.1 Background and motivation	1
1.2 Surrogate-driven inverse design: a generalizable framework	2
1.3 Problem structuring and justification	2
1.3.1 Gyroid unit cell: shape and mathematical representation	3
1.3.2 Objective: maximise surface area	3
1.3.3 Design constraints and their physical significance	3
1.4 Project objectives	4
2 Literature Review	5
2.1 Surrogate modelling in engineering design	5
2.2 Design optimisation techniques	6
2.3 Surrogate-assisted inverse design	6
2.4 Gyroid heat exchangers: state-of-the-art	6
3 Formulation and Methodology	7
3.1 Problem formulation	7
3.2 Simulation Methodology: Lattice Boltzmann Method	8
3.2.1 Discretization and Boundary Conditions	8
3.3 Data Generation Strategy	10
3.4 Physics-Guided Surrogate Model Architecture	11
3.5 Optimisation Strategy and Pipeline	12
3.5.1 Optimisation Algorithms Compared	12
3.5.2 Pipeline Comparison and Final Strategy	12
3.5.3 Final Hybrid Optimisation Pipeline	13
4 Results and Discussion	14
4.1 Surrogate Model Performance	14
4.2 Optimisation Results	15

4.3	Comparative Analysis: Mid-Semester vs. Final Methodology	16
4.3.1	Methodology Comparison	16
4.3.2	Results Comparison	17
5	Summary and Conclusions	19

List of Tables

4.1	Performance of GPR Surrogate Models on the Test Set.	14
4.2	Optimal Geometric Design Parameters (from Local Refinement).	16
4.3	Comparison of Optimal Design from Mid-Semester and Final Report. . .	17

List of Figures

3.1	Grid convergence analysis showing Average Velocity (v_{avg}) stabilising as cell size is refined. Convergence occurs at a cell size of 0.5.	9
3.2	The two-stage, physics-informed surrogate model architecture. Model 1 predicts geometric properties, which are then fed into Model 2 along with flow conditions to predict fluid dynamic properties.	11
4.1	Contour plot of predicted mass (filled) and surface area (dashed lines) within the trust region [14-16, 24-25]. The red line indicates the 125 g mass constraint.	15
4.2	Flowchart depicting the optimization pipeline.	18

Notation and Abbreviations

C_x Lattice cell size in the X-direction (mm)

C_{yz} Lattice cell size in the Y/Z-directions (mm)

v_{in} Inlet flow velocity (mm/s)

v_{avg} Average flow velocity in the core (mm/s)

Δp Pressure drop across the heat exchanger (Pa)

A_s Core surface area (mm²)

M Mass of the heat exchanger core (g)

D_h Hydraulic diameter (m)

Re Reynolds number (dimensionless)

Nu Nusselt number (dimensionless)

ANN Artificial Neural Network

GPR Gaussian Process Regressor

CFD Computational Fluid Dynamics

LBM Lattice Boltzmann Method

LHS Latin Hypercube Sampling

SMBDO Surrogate-Modelling-Based Design Optimisation

RMSE Root Mean Squared Error

TPMS Triply Periodic Minimal Surface

Chapter 1

Introduction

1.1 Background and motivation

Modern engineering design across aerospace, automotive, and biomedical domains is becoming ever more complex and multi-objective. Achieving higher performance under tight constraints on mass, volume, and energy requires accurate prediction of physics such as fluid flow, structural response, and heat transfer. These behaviors are governed by nonlinear partial differential equations and are typically evaluated using high-fidelity numerical methods such as Computational Fluid Dynamics (CFD) and Finite Element Analysis (FEA) (Versteeg and Malalasekera, 2007).

Relying on high-fidelity simulations creates a practical bottleneck: single runs can take hours or days on HPC systems, which conflicts with the iterative nature of design workflows (Bhaskar and Roy, 2017). Full optimisation, sensitivity studies, and design-space exploration demand many function evaluations; when each evaluation is expensive, an optimisation can become infeasible in practice.

Additive Manufacturing (AM) expands design freedom by allowing highly complex internal geometries, such as lattice or TPMS (Triply Periodic Minimal Surface) structures, that were previously impossible or impractical. That increased freedom is a double-edged sword: it enables superior performance but also vastly enlarges the design space, making brute-force or manually guided optimisation impractical. The result is a pressing need

for methods that retain the benefits of high-fidelity physics models while dramatically reducing the cost per evaluation.

1.2 Surrogate-driven inverse design: a generalizable framework

To resolve this tension between design freedom and computational cost, we adopt a surrogate-model-driven inverse design framework. A surrogate (or metamodel) is a data-driven function that approximates the mapping from design variables to performance metrics. The typical workflow is: run a set of strategically chosen high-fidelity simulations, train a machine-learning model (e.g., a GPR) on that data, and then use the trained surrogate to predict performance for new designs at negligible cost (Forrester et al., 2008; Bhaskar and Roy, 2017).

The main advantage is computational efficiency: the expensive CFD work is performed offline to create the training dataset; once the surrogate is trained, predictions are orders of magnitude faster than running full simulations. This makes inverse design—where one specifies desired performance and uses optimisation to find the design parameters that achieve it—tractable.

This general approach is domain-agnostic. For this project, we apply it to heat exchanger design because heat exchangers are widely used and improvements can yield substantial energy and weight savings. The framework’s modular components can be transferred to other physics problems.

1.3 Problem structuring and justification

We focus on developing and demonstrating a surrogate-assisted inverse design workflow for a gyroid heat exchanger. The inverse design problem is to find geometric and flow parameters that maximise a performance metric while satisfying engineering constraints.

1.3.1 Gyroid unit cell: shape and mathematical representation

The gyroid belongs to the TPMS family: a smooth, periodic surface without planar symmetry or straight lines. Its topology yields a high surface-to-volume ratio and continuous, tortuous channels that are beneficial for heat transfer and mixing. The gyroid's implicit equation is:

$$\sin\left(\frac{2\pi x}{C_x}\right)\cos\left(\frac{2\pi y}{C_{yz}}\right) + \sin\left(\frac{2\pi y}{C_{yz}}\right)\cos\left(\frac{2\pi z}{C_{yz}}\right) + \sin\left(\frac{2\pi z}{C_{yz}}\right)\cos\left(\frac{2\pi x}{C_x}\right) = 0$$

Here C_x and C_{yz} control the lattice spacing in the x and y/z directions respectively. The zero level set defines the surface: positive and negative values define the two complementary volumes separated by the surface. The connected channels avoid dead-ends, reducing flow maldistribution and aiding uniform heat transfer. By varying C_x and C_{yz} we directly affect the core's surface area, mass, and the geometry of the flow passages—variables that are central to the optimisation problem.

1.3.2 Objective: maximise surface area

The goal is to maximise the core surface area A_s . For a heat exchanger, the total heat transfer rate is

$$Q = UA_s\Delta T_{LM}$$

where U is the overall heat transfer coefficient and ΔT_{LM} is the log-mean temperature difference (Incropera et al., 2007). For given fluids and operating conditions, U and ΔT_{LM} are approximately fixed, so increasing A_s is the most direct lever to increase heat transfer capacity in a fixed envelope.

1.3.3 Design constraints and their physical significance

To ensure the design is practical and realistic, we impose the following constraints:

- **Mass ($M < 125$ g):** Many applications, especially mobile and airborne systems, are mass-sensitive. A hard mass limit prevents impractically heavy cores.

- **Pressure Drop ($\Delta p < 8000 \text{ Pa}$):** Pressure drop impacts pumping power ($P_{pump} \propto \Delta p \cdot \dot{V}$) (White, 2011). Excessive pressure loss increases parasitic power and reduces system efficiency.
- **Average Velocity ($v_{avg} > 520 \text{ mm/s}$):** A minimum mean flow velocity helps maintain convective heat transfer effectiveness and avoid stagnant regions and thick thermal boundary layers (Incropera et al., 2007).

1.4 Project objectives

The primary goals for this work are:

1. Develop a computationally cheap surrogate that predicts the four key outputs (M , A_s , Δp , v_{avg}) from the three inputs (C_x , C_{yz} , v_{in}).
2. Structure the surrogate to reflect relevant physics to improve accuracy and generalisability.
3. Implement an efficient data generation and optimisation routine that exploits the surrogate's speed to solve the constrained inverse design problem.

Chapter 2

Literature Review

This chapter reviews background work that underpins our approach. We cover surrogate modelling, optimisation techniques, surrogate-assisted inverse design, and recent work on gyroid-based heat exchangers.

2.1 Surrogate modelling in engineering design

Because high-fidelity simulations are expensive, surrogate models (metamodels) are widely used to approximate costly codes (Forrester et al., 2008). Surrogates learn input-output mappings from a limited set of simulations. Methods range from polynomial response surfaces and Kriging to Radial Basis Functions (RBFs) and Artificial Neural Networks (ANNs) (Bhaskar and Roy, 2017). Gaussian Process Regressors (GPRs), a form of Kriging, are particularly powerful as they provide not only a prediction but also a measure of uncertainty. ANNs are particularly useful for highly nonlinear, high-dimensional problems and have been applied in thermofluid problems, often yielding large speed-ups over CFD.

2.2 Design optimisation techniques

Optimisation algorithms can be gradient-based or gradient-free. Gradient-based methods are efficient but can become trapped in local optima; gradient-free approaches (e.g., genetic algorithms) explore globally but often need many evaluations (Arora, 2011). Bayesian Optimisation offers a balanced alternative by building a probabilistic surrogate (like a GPR) and using an acquisition function to choose informative samples, making it effective for expensive black-box functions (Shahriari et al., 2015).

2.3 Surrogate-assisted inverse design

Inverse design specifies desired performance and searches for the design that achieves it. When each simulation is expensive, inverse design becomes impractical; replacing the simulator with a fast surrogate makes the optimisation loop tractable. A practical strategy is to use a global search method (e.g., Bayesian Optimisation) on the surrogate, then refine promising solutions with a local method.

2.4 Gyroid heat exchangers: state-of-the-art

TPMS structures like the gyroid are attractive for heat exchangers because of their high surface-area-to-volume ratios and complex internal channels. (Kus et al., 2024) performed numerical and experimental studies on LPBF-manufactured gyroid heat exchangers and validated CFD against experiments, finding that cell size strongly controls the trade-off between heat-transfer area and pressure drop. Smaller cells boost surface area but sharply increase pressure loss—a key trade-off our optimisation must manage. Other studies agree that TPMS geometries promote mixing and enhance convective heat transfer.

Chapter 3

Formulation and Methodology

3.1 Problem formulation

We seek the design parameters that maximise the core surface area A_s . The design vector is (C_x, C_{yz}, v_{in}) . Formally:

$$\arg \max_{C_x, C_{yz}, v_{in}} A_s(C_x, C_{yz})$$

subject to:

$$M(C_x, C_{yz}) < 125 \text{ g} \quad (3.2)$$

$$\Delta p(C_x, C_{yz}, v_{in}) < 8000 \text{ Pa} \quad (3.3)$$

$$v_{avg}(C_x, C_{yz}, v_{in}) > 520 \text{ mm/s} \quad (3.4)$$

and bounds:

$$10 \text{ mm} \leq C_x \leq 25 \text{ mm} \quad (3.5)$$

$$10 \text{ mm} \leq C_{yz} \leq 25 \text{ mm} \quad (3.6)$$

$$2500 \text{ mm/s} \leq v_{in} \leq 3500 \text{ mm/s} \quad (3.7)$$

The mappings $A_s(\cdot)$, $M(\cdot)$, $\Delta p(\cdot)$, and $v_{avg}(\cdot)$ are approximated by the surrogate models.

3.2 Simulation Methodology: Lattice Boltzmann Method

For the CFD work, we used a Lattice Boltzmann Method (LBM) solver. Unlike conventional approaches that solve the macroscopic Navier–Stokes equations, LBM operates at a **mesoscopic level** through the Lattice Boltzmann Equation (LBE) (Chen and Doolen, 1998). The fluid is treated as a set of particle distribution functions f_i defined on a discrete lattice.

The dynamics follow a discretized form of the Boltzmann equation:

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) - f_i(\mathbf{x}, t) = \Omega_i(f)$$

Each time step consists of two operations:

1. **Stream:** The distributions f_i at position $\mathbf{x}$ move to neighbouring nodes along the discrete velocities $\mathbf{c}_i$.
2. **Collide:** Momentum is redistributed locally through the collision operator $\Omega_i(f)$.

We used the standard BGK model:

$$\Omega_i(f) = -\frac{1}{\tau} [f_i(\mathbf{x}, t) - f_i^{eq}(\mathbf{x}, t)]$$

The relaxation time τ controls how quickly the distributions approach equilibrium and is directly tied to the fluid’s kinematic viscosity ν , which in turn fixes the Reynolds number for the simulation.

Macroscopic quantities are reconstructed from the distributions at every step:

$$\rho = \sum_i f_i, \quad \rho \mathbf{u} = \sum_i f_i \mathbf{c}_i$$

3.2.1 Discretization and Boundary Conditions

LBM uses a uniform Cartesian (**voxel**) grid, and the gyroid geometry is simply embedded into it. This changes how wall boundaries are applied. For no-slip walls, we used the

bounce-back approach (Chen and Doolen, 1998): any distribution streaming into a wall node is reflected straight back, enforcing zero velocity at the wall without additional modelling.

Our flow regime ($Re \approx 700$) is laminar, so the boundary layer is thick and strongly influenced by viscous diffusion (White, 2011; Incropera et al., 2007). This plays to LBM's strengths, since the method resolves the boundary-layer behaviour directly at the voxel level.

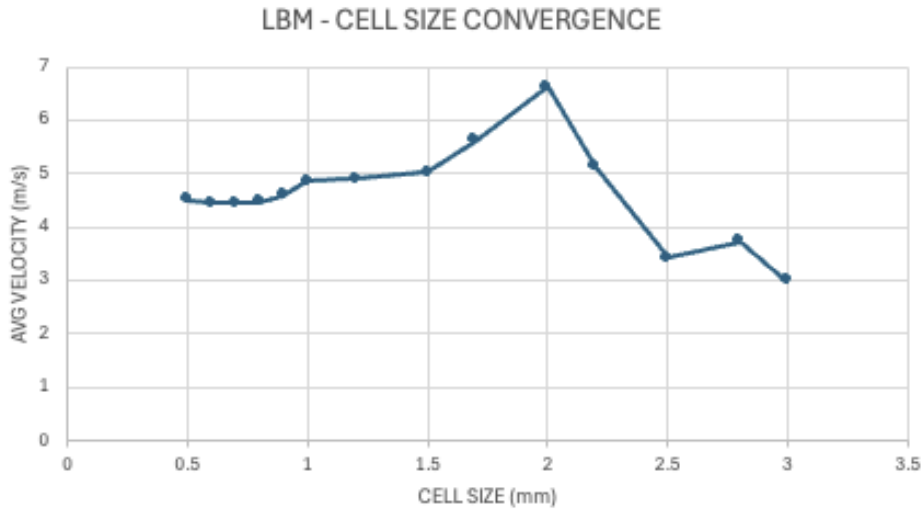


Figure 3.1: Grid convergence analysis showing Average Velocity (v_{avg}) stabilising as cell size is refined. Convergence occurs at a cell size of 0.5.

A grid convergence check was performed to confirm that the results were not dependent on grid resolution. We ran a representative geometry while progressively refining the voxel size and tracked the average velocity v_{avg} .

Figure 4.2 shows the expected asymptotic behaviour: as the voxel size decreased, the predicted v_{avg} value approached a stable limit. Convergence was effectively reached at a cell size of **0.5**, which we therefore used for all subsequent simulations as a reasonable compromise between accuracy and computational cost.

3.3 Data Generation Strategy

We used a mixed approach to build the dataset. It started with a broad set of initial samples, and then we kept adding points in a more targeted way as the model learned.

- **Initial Dataset:** We first created 75 samples using **Latin Hypercube Sampling (LHS)** across the full input range (X, YZ: 10–25 mm). This gave us a reasonably spread-out starting set to work from.
- **Active Learning:** After that, the dataset grew through an iterative active learning loop. Each cycle trained Gaussian Process Regression (GPR) models on whatever data we had at the time (Shahriari et al., 2015), and an acquisition rule picked the next points that seemed most worthwhile to simulate.
- **Acquisition Function:** The custom acquisition setup balanced exploration and exploitation, but not in an overly rigid way. Two pieces drove the scoring:
 1. *Relevance (Exploitation):* Points close to the 125 g mass limit were favored, since the best designs are expected to land right along that boundary. This part of the score pushed sampling toward that region of interest.
 2. *Uncertainty (Exploration):* We also included the normalized standard deviations from both the mass and surface-area GPR models. This kept the loop from getting stuck in areas where the model “looked” confident but actually had little support.

Multiplying these two terms helped the algorithm focus on points that were near the constraint and also carried meaningful uncertainty. Each iteration selected 10 new candidates, and after 20 rounds, all simulated points were added into a single consolidated dataset.

3.4 Physics-Guided Surrogate Model Architecture

The Artificial Neural Networks (ANNs) from the mid-term report were replaced with **Gaussian Process Regressors (GPRs)** (Forrester et al., 2008). GPRs were chosen for their ability to provide robust uncertainty estimates, which is essential for the active learning acquisition function.

The physics-guided two-stage architecture, which proved successful in the mid-term, was retained (Forrester et al., 2008). This architecture (shown in Figure 3.2) decouples the geometric and fluid-dependent predictions.

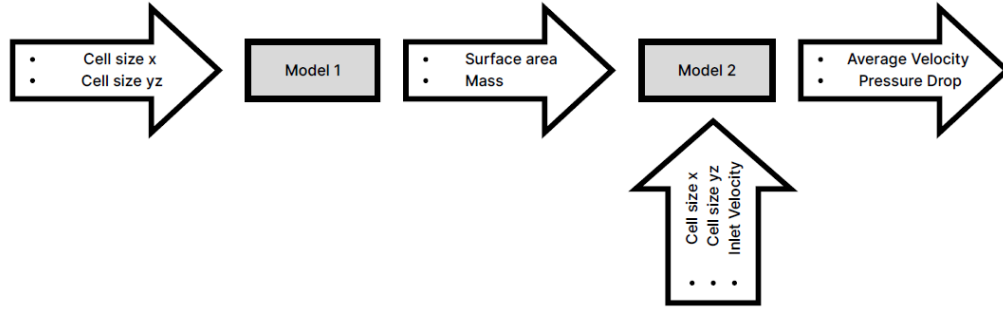


Figure 3.2: The two-stage, physics-informed surrogate model architecture. Model 1 predicts geometric properties, which are then fed into Model 2 along with flow conditions to predict fluid dynamic properties.

- **Model 1 (Geometric Predictor):** A GPR model that takes geometric parameters (C_x, C_{yz}) as input and predicts the geometric properties (M, A_s) (Forrester et al., 2008).
- **Model 2 (Fluid Dynamics Predictor):** A GPR model that takes the geometric parameters (C_x, C_{yz}), the inlet velocity (v_{in}), and the *predicted* outputs from Model 1 (M, A_s) as inputs to predict the fluid dynamic properties ($\Delta p, v_{avg}$) (Forrester et al., 2008).

This staged design embeds physical intuition: fluid dynamics are dependent not just on the primary inputs, but on the resulting geometric properties like mass and wetted surface area (Incropera et al., 2007).

3.5 Optimisation Strategy and Pipeline

The optimisation relied on the trained GPR surrogate models, which are extremely quick to evaluate. We compared two main ways of running the overall optimisation: one built around Bayesian Optimisation and another using a gradient-based search. After the main search, both approaches ended with the same local refinement steps using a small grid and a trust-region method.

3.5.1 Optimisation Algorithms Compared

Bayesian Optimisation (BO)

Bayesian Optimisation works well when evaluating a function is expensive, which fits our situation since the surrogates come from limited CFD data (Shahriari et al., 2015). It keeps a probabilistic model of the function (the GPR), which gives both a mean estimate and an uncertainty. Using an acquisition rule, it tries to balance checking regions that look promising with checking areas where the model is still unsure. This makes the global search efficient without needing a huge number of evaluations.

Gradient-Based Optimisation (GBO)

Gradient-Based Optimisation (GBO) moves locally by following the steepest direction of improvement based on the surrogate's derivatives (Arora, 2011). It is quick and usually converges in only a few steps. The tradeoff is that, unless the starting point is already near the global best, it can easily settle into a nearby local optimum.

3.5.2 Pipeline Comparison and Final Strategy

We tried two versions of the pipeline, each followed by the same local refinement step:

- **Pipeline 1 (BO-based):** The main search used Bayesian Optimisation.
- **Pipeline 2 (GBO-based):** The main search used Gradient-Based Optimisation.

Both pipelines pointed to the same region in the design space and produced nearly identical results. The Bayesian Optimisation run, however, reached a slightly better objective (a bit more surface area), mainly because it handled local peaks more reliably. For this reason, the BO-based workflow was chosen as the final approach.

3.5.3 Final Hybrid Optimisation Pipeline

The final setup used a three-stage process built around the BO search:

1. **Stage 1: Global Exploration.** We first ran the **Bayesian Optimisation** routine (`constrained_bayesian_opt`) with GPR Model 1 across the full space (10–25 mm in both C_x and C_{yz}) (Forrester et al., 2008). This gave a clear region where A_s is maximized while still meeting the $M < 125$ g constraint. The promising area fell around $C_x \in [14, 16]$ and $C_{yz} \in [24, 25]$.
2. **Stage 2: Trust Region Refinement.** Using that region as a guide, we set up a tighter trust region and sampled **50 new points** with LHS inside those bounds. These new samples were added to the dataset, and the GPR models were retrained.
3. **Stage 3: Final Local Optimisation.** We then ran another BO step along with a small grid-based refinement, but this time restricted to the trust region. This produced a more accurate, locally validated optimum.
4. **Stage 4: Feasibility Search.** The optimal pair (C_x, C_{yz}) was fixed for Model 2, and a simple 1D sweep over v_{in} determined which inlet velocities satisfy both the Δp and v_{avg} constraints.

Chapter 4

Results and Discussion

4.1 Surrogate Model Performance

The GPR models were checked against a held-out test split (10% of the original LHS samples). Their performance is shown in Table 4.1. Overall, the surrogates captured the main behaviour of the system, but the accuracy varied across outputs. The geometric parts of Model 1 performed reliably, while the mass prediction in Model 1 and the pressure-drop prediction in Model 2 had noticeably lower R^2 values. This hinted at some overfitting in the early stages, which was later reduced by adding points through Active Learning and the Trust Region sampling.

Table 4.1: Performance of GPR Surrogate Models on the Test Set.

Model 1 (Geometric)		
Mass (g)	0.239	0.798
Surface Area (mm ²)	117.515	0.933
Model 2 (Fluid Dynamics)		
Avg. Velocity (mm/s)	17.680	0.925
Pressure Drop (Pa)	1084.186	0.686

4.2 Optimisation Results

The three-step optimisation framework produced the final geometric settings listed in Table 4.2. The optimum lands almost exactly on the 125 g mass boundary, which matches the physical expectation that designs with larger surface area usually sit right at the upper mass limit.

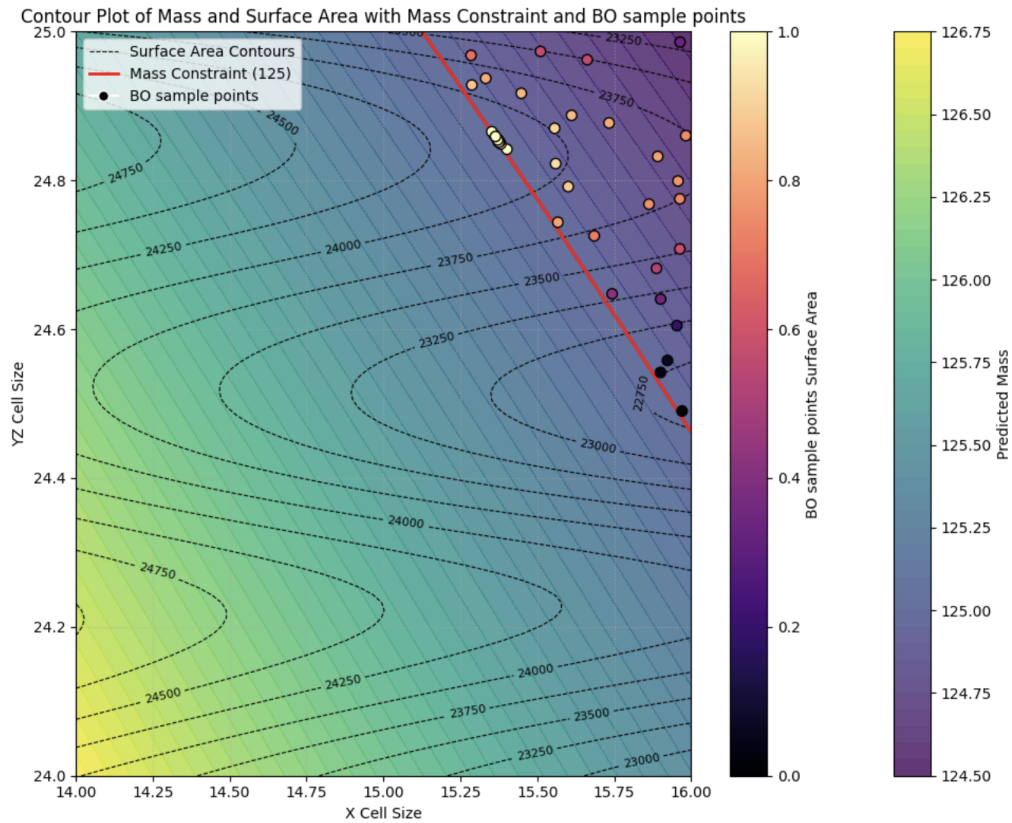


Figure 4.1: Contour plot of predicted mass (filled) and surface area (dashed lines) within the trust region [14-16, 24-25]. The red line indicates the 125 g mass constraint.

Figure 4.1 shows the contour map from the updated GPR Model 1 inside the trust region. Filled contours correspond to predicted mass, dashed lines to surface area, and the red curve marks the 125 g constraint. The chosen optimum lies directly on that boundary.

Table 4.2: Optimal Geometric Design Parameters (from Local Refinement).

Parameter	Optimal Value	Constraint / Target
Inputs (Design Variables)		
Cell Size $X(C_x)$	15.3645 mm	[14, 16] mm
Cell Size $YZ(C_{yz})$	24.8605 mm	[24, 25] mm
Outputs (Performance)		
Surface Area (A_s)	24116.76 mm ²	Maximised
Mass (M)	124.9967 g	< 125 g

4.3 Comparative Analysis: Mid-Semester vs. Final Methodology

The final workflow differs substantially from the mid-semester version, especially in how data was gathered, how the surrogate was built, and how the optimisation was carried out. These changes produced a more stable, data-efficient process and a noticeably better design.

4.3.1 Methodology Comparison

- **Data Generation Strategy:** The mid-semester submission relied on a **Full-Factorial Design of Experiments**, which grows quickly in cost as variables increase. In contrast, the final approach used a mixed strategy and needed only **319 simulations** in total. The dataset was assembled in three phases, gradually focusing on the most relevant region.
- **Surrogate Model:** Earlier work used Artificial Neural Networks (ANNs) These were replaced with **Gaussian Process Regressors (GPRs)** (Forrester et al., 2008), mainly because GPRs provide uncertainty estimates that drive the Active Learning step. The two-part structure separating geometric and fluid-dynamic predictions stayed the same.
- **Optimisation Strategy:** The mid-semester pipeline combined a global Bayesian search with a local gradient method (Arora, 2011). The final work expanded this

into a multi-stage **trust-region surrogate optimisation** process, where the surrogate was retrained with additional samples near the promising region before running the final refinement.

4.3.2 Results Comparison

The updated methodology reduced the number of simulations and still produced a better optimum, summarised in Table 4.3.

Table 4.3: Comparison of Optimal Design from Mid-Semester and Final Report.

Parameter	Mid-Semester (ANN)	Final (GPR + Active Learning)
Data Generation	Full-Factorial DoE	LHS + Active Learning
Data Points Used	787	319
C_x (mm)	21.787	15.3645
C_{yz} (mm)	20.529	24.8605
Mass (M)	124.99 g	124.9967 g
Surface Area (A_s)	23559.94 mm²	24116.76 mm²

The final optimisation produced a surface area of 24,116.76 mm², which is roughly 557 mm² (about 2.36%) higher than the mid-semester value of 23,559.94 mm². This outcome highlights the benefit of pairing Active Learning with a trust-region refinement step: the improved surrogate required far fewer simulations while still locating a better region of the design space than the full-factorial setup.

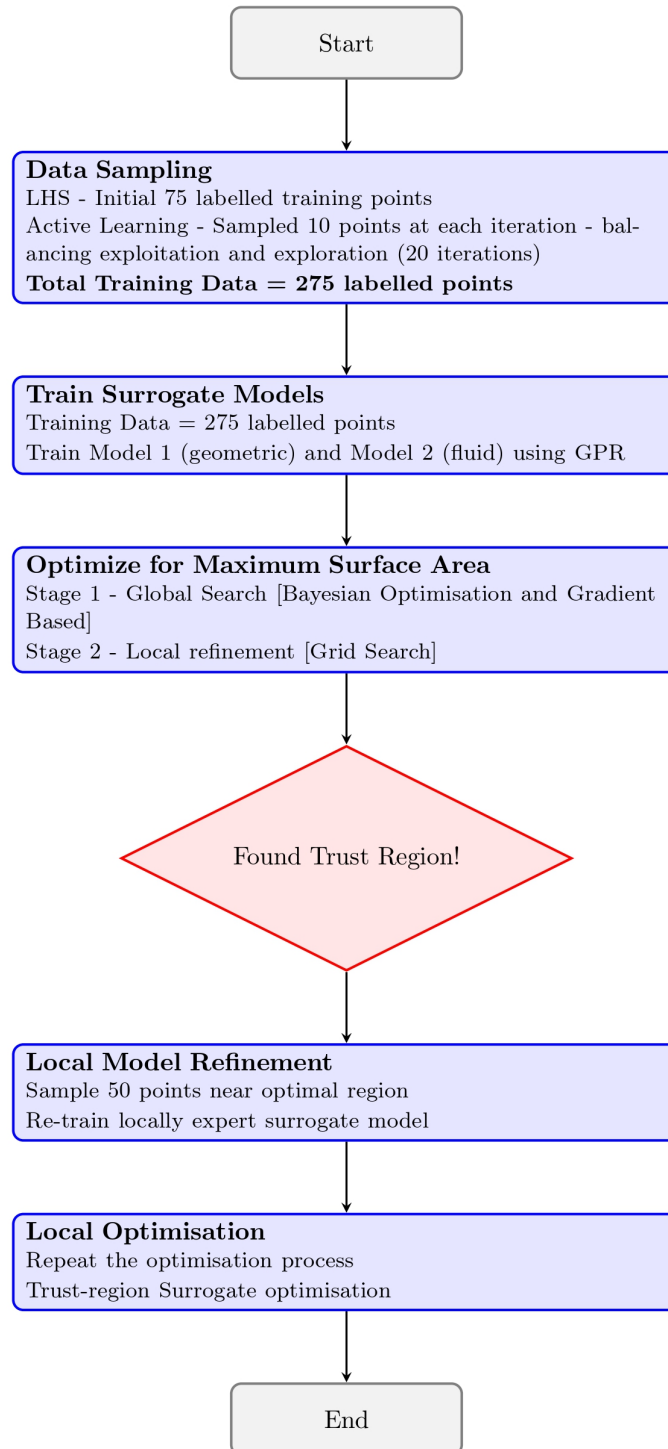


Figure 4.2: Flowchart depicting the optimization pipeline.

Chapter 5

Summary and Conclusions

We built and applied a surrogate-driven inverse-design framework for a gyroid heat-exchanger geometry. The goal was to maximise surface area while meeting limits on mass, pressure drop, and mean flow velocity.

The surrogate consisted of two Gaussian Process Regression models, one handling the geometry-derived quantities and the other predicting flow-related outputs. Data was generated in two stages: an initial LHS sweep, followed by an active-learning phase that selected new samples based on relevance to the mass boundary and on model uncertainty.

The optimisation followed a three-part process: a broad Bayesian search, a trust-region stage with added samples, and a final local refinement. This sequence converged to a design at $(C_x, C_{yz}) \approx (15.36 \text{ mm}, 24.86 \text{ mm})$, yielding a predicted surface area near $24,117 \text{ mm}^2$ while landing exactly on the 125 g mass limit. The workflow reduced the design effort from what would normally require several weeks of simulation to a workflow manageable within hours.

References

- Arora, J. S. (2011). *Introduction to optimum design*. Academic press.
- Bhaskar, A. and Roy, S. (2017). Prospects of surrogate modeling for the design of engineering systems. *Journal of the Indian Institute of Science*, 97:477–494.
- Chen, S. and Doolen, G. D. (1998). Lattice boltzmann method for fluid flows. *Annual Review of Fluid Mechanics*, 30(1):329–364.
- Forrester, A. I., Sobester, A., and Keane, A. J. (2008). *Engineering design via surrogate modelling: a practical guide*. John Wiley & Sons.
- Incropera, F. P., DeWitt, D. P., Bergman, T. L., and Lavine, A. S. (2007). *Fundamentals of heat and mass transfer*. John Wiley & Sons.
- Kus, A., Fijałkowski, M., Zaborowska, E., Pasko, J., and Ochrymiuk, T. (2024). Numerical and experimental investigation of a gyroid heat exchanger manufactured via lpb. *International Journal of Heat and Mass Transfer*, 223:125203.
- Shahriari, B., Swersky, K., Wang, Z., Adams, R. P., and De Freitas, N. (2015). Taking the human out of the loop: A review of bayesian optimization. *Proceedings of the IEEE*, 104(1):148–175.
- Versteeg, H. K. and Malalasekera, W. (2007). *An introduction to computational fluid dynamics: the finite volume method*. Pearson Education.
- White, F. M. (2011). *Fluid mechanics*. McGraw-Hill.